

Benchmark	base → ODILE					
	Llama-3.1-8B	Qwen-2.5-7B	Qwen-2.5-14B	Qwen3-8B*	Qwen3-32B [#]	Llama-3.3-70B
AlpacaEval LC ($n=805$)	32.0 → 31.6	38.7 → 38.2	44.8 → 35.5	19.06 → 21.14	79.83 → 61.35	48.6 → 47.9
GSM8K ($n=200$)	85.5 → 83.5	89.0 → 87.5	91.5 → 90.5	95.5 → 94.5	97.5 → 71.5	95.0 → 95.5
IFEval strict ($n=200$)	62.5 → 59.5	54.5 → 51.5	64.0 → 52.0	64.5 → 57.0	66.5 → 27.0	73.5 → 71.0
BFCL non-live AST [†] ($n=1,240$)	73.98 → 73.35	74.98 → 75.12	—	—	—	—
τ -bench airline [◇] ($n=50$)	42.00 → 38.78	—	—	—	—	—

Table 5: **General-purpose capability under ODILE.** Each cell shows base → ODILE. On AlpacaEval / GSM8K / IFEval, ODILE preserves base capability within ± 7.5 pp on Llama-3.1-8B, Qwen-2.5-7B, Llama-3.3-70B, and Qwen3-8B; Qwen-2.5-14B shows a moderate AlpacaEval LC (−9 pp) and IFEval (−12 pp) drop attributable to its long-form generation under the orth objective (Appendix ??). Qwen3-8B is the only backbone where ODILE *improves* AlpacaEval LC (+2 pp); the likely mechanism is a mild ODILE-induced output-shortening (6293 → 6176 chars on average) that benefits the length-controlled win rate. **Qwen3-32B is the high-benign-cost backbone:** ODILE’s representation-level intervention over-jams on benign instruction-format-constrained inputs, with measurable drops on IFEval (−39.5 pp), GSM8K (−26 pp), and AlpacaEval LC (−18.5 pp); the model jams 9999... on $\sim 98/200$ benign IFEval prompts and $\sim 20/200$ benign GSM8K prompts. This is a genuine ODILE over-firing on benign at 32B scale, distinct from the thinking-mode *measurement* artifacts that previously inflated the Qwen3 deltas (see [#] below and Appendix ??). **Honest takeaway:** ODILE’s benign cost scales with model size — at 8B (Llama, Qwen3) the cost is near-zero or positive; at Qwen-2.5-14B and Qwen3-32B the cost is measurable on long-form / instruction-constrained inputs; at 70B-Llama the cost is again within noise. *Qwen3-8B AlpacaEval LC uses the `weighted_alpaca_eval_gpt4.1` annotator (gpt-4.1-2025-04-14); Llama-3.1-8B / Qwen-2.5-7B / Qwen-2.5-14B / Llama-3.3-70B use a local Llama-3.3-70B judge. Absolute LC values are not directly comparable across the two judges, though the sign of the LC delta is preserved in spot checks. [#]Qwen3-32B cells are *corrected, no-think* re-runs (2026-05-23). Qwen3 emits a `<think>...</think>` block before its final answer; under the harnesses’ default token budgets and judge configurations this caused: (a) truncation mid-`<think>` when the budget was too low (e.g., the IFEval verifier scored the reasoning text against format constraints, 200/200 unparseable); (b) judges / classifiers reading the `<think>` block (e.g., the TensorTrust judge mis-bucketed coherent answers as `B_leaked` whenever the access code appeared in the thinking trace); (c) AlpacaEval length-control penalizing the `<think>` verbosity (avg_length = 6259 chars on Qwen3-32B base under default thinking). We re-ran IFEval, GSM8K, AlpacaEval, and TT-benign with `enable_thinking=False` (which corrects the artifact for these parser-strict benchmarks) and re-judged TT-benign with gpt-4o-mini. AlpacaEval was also re-judged with gpt-4o-mini under the no-think regeneration. The corrected base capabilities are high and on-par with same-scale peers (IFEval 66.5, GSM8K 97.5, AlpacaEval LC 79.83); only the ODILE-side cost is real over-firing. Full methodology + per-metric provenance + script paths in Appendix ??.

[†]BFCL [ref] v4 non-live AST aggregated over the five subsets `simple_python`, `parallel`, `multiple`, `parallel_multiple`, and `irrelevance` ($n=1,240$ per backbone-defense cell). Two backbones populated for the rebuttal window; remaining four backbones pending on warm-endpoint availability. Per-subset breakdowns in Appendix ??.

[◇] τ -bench [ref] airline-domain base split ($n=50$ tasks, 1 trial; average reward in $[0, 100]$). ODILE preserves agentic multi-turn tool-use to within −3.2 pp of base on Llama-3.1-8B; one `infrastructure_error` on the ODILE side stems from 32K context-window pressure, addressable by raising context to 65K (Appendix ??). Agentic benign-utility cells under no-attack live in Tables ??, ??, ??.

Canonical cell aggregator: `paper/latestresults/session_2026-05-20/cross_model/SUMMARY.md` §§1f, 1h, 1i.